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# The Effect of Polyunsaturated Fats on Bile Acid Metabolism and Cholelithiasis in Squirrel Monkeys

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**Squirrel monkeys (*Saimiri sciureus*) were fed diets containing safflower oil, butter, or coconut oil and 1 mg cholesterol/cal for 15–17 mo to examine the effect of type of fat on cholelithiasis and bile acid metabolism. Controls were fed low cholesterol diets containing an isocaloric mixture of the three fats. Cholic acid fractional catabolic rate, pool size, chenodeoxycholic acid pool size, and total bile acid pool size and excretion rate were estimated using a modification of Lindstedt's isotopic turnover procedure.**

**The animals fed the safflower oil diet had the highest incidence of cholelithiasis (9/10) when compared to those fed butter (3/7) and coconut oil (1/7). Animals consuming the low cholesterol control diet did not develop gallstones. The butter- and coconut oil-fed groups had significantly ($p < 0.05$) expanded bile acid pools when compared to controls, and the butter-fed group had a significantly increased ($p < 0.05$) cholic acid fractional catabolic rate. The safflower oil group had the smallest mean bile acid pool and the highest mean lithogenic index of the cholesterol-fed groups.**

**It was concluded that the safflower oil-fed animals had a higher incidence of cholelithiasis than the butter group because, unlike the latter group, they did not compensate for a high cholesterol intake by stimulating bile acid synthesis. The animals consuming coconut oil apparently did not absorb cholesterol to the extent of the other groups and as a result their bile did not become saturated with cholesterol.**

CHOLESTEROL IS MAINTAINED in solution in the bile by bile salts and lecithin in the form of a mixed micelle. When the secretion of cholesterol exceeds the normal solubilizing capacity of the bile salt-lecithin micelle, the liver

*Deceased.

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produces micelles that are supersaturated with cholesterol. These supersaturated micelles are relatively unstable and eventually release the excess cholesterol as insoluble crystals. The crystals are thought to adhere to particulate matter in the bile and thereby create the nidus for gallstone formation.[1]

The mechanism(s) responsible for the production of supersaturated micelles has not been determined; however, any factor that stimulates the secretion of cholesterol by the liver is potentially lithogenic. As a result, hypocholesterolemic agents are being reexamined with regard to their effect on bile composition. Among those of interest are polyunsaturated fats, for recent evidence suggests that high dietary levels of this type of fat may stimulate the production of gallstones.[2,3]

The present study was designed to evaluate the effect of polyunsaturated fats on the process of gallstone formation in the squirrel monkey and to determine the mechanism of action. The squirrel monkey was chosen because the disease is relatively easy to induce in this species by the addition of cholesterol to the diet and because sterol metabolism in the squirrel monkey has been rather thoroughly evaluated. Gallstone formation usually occurs within several months after the initiation of cholesterol feeding and generally the stones appear similar to those found in man.[4]

## MATERIALS AND METHODS

### *Animals and Diets*

Mature Brazilian squirrel monkeys (*Saimiri sciureus*) obtained from the Tarpon Zoo, Tarpon Springs, Fla., were assigned to 1 of 4 groups. Each group was fed a diet high in polyunsaturated fat (safflower oil), saturated fat (butter or coconut oil), or a mixed fat diet (Table 1). The animals were weighed, and plasma cholesterol concentrations were determined monthly using the AutoAnalyzer II methodology of Rush et al.[5] The animals were fed the experimental diets for 15–17 mo before the bile acid turnover measurements were started. Each group contained roughly equal numbers of males and females.

**Table 1. Composition of Diets**

| | g/100 g | | | |
|---|---|---|---|---|
| Ingredients | Safflower Oil | Butter | Coconut Oil | Mixed Fat |
| Casein | 12.6 | 12.6 | 12.6 | 12.6 |
| Dry milk solids | 29.1 | 29.1 | 29.1 | 29.1 |
| Flour | 19.4 | 19.4 | 19.4 | 19.4 |
| Applesauce | 7.1 | 7.1 | 7.1 | 7.1 |
| Safflower oil | 22.0 | — | 2.0 | 7.2 |
| Butter | — | 27.2 | — | 9.1 |
| Coconut oil | — | — | 19.7 | 7.3 |
| Crystalline cholesterol | 0.44 | 0.35* | 0.44 | — |
| Vitamin mix | 2.1 | 2.1 | 2.1 | 2.1 |
| Vitamin $D_3$ | 240 IU | 240 IU | 240 IU | 240 IU |
| Salts XIV | 1.9 | 1.9 | 1.9 | 1.9 |
| Alphacel | 5.1 | — | 5.5 | 4.0 |
| $\beta$-Sitosterol | — | 0.7† | — | — |

All diets contained 45% of calories as fat, 25% of calories as protein, and 30% of calories as carbohydrate. The diets contained 435 Cal/100 g and 1.0 mg cholesterol/Cal.

*Part of the cholesterol in this diet was derived from the butter, thus less crystalline cholesterol was needed.

†The $\beta$-sitosterol was added to the butter diet in order to equalize the $\beta$-Sitosterol content of all diets.

### *Isotope Preparation and Administration*

Each animal was injected with approximately 50 $\mu$Ci of cholic acid-24-$^{14}$C (New England Nuclear, Boston, Mass.). The purity of the isotope exceeded 95%; nonetheless, it was repurified just prior to injection by thin-layer chromatography on silica gel G using ethyl acetate:chloroform:acetic acid (45:45:10) as the developing solvent. The isotope was dissolved in 0.4 ml of ethanol, diluted with saline, and injected into the saphenous vein.

### *Feces and Bile Collection*

After the isotope was injected, the animals were placed in individual sterol balance cages and complete fecal collections were made each day for 5 days. Feces were collected in preweighed screw-cap jars and stored frozen for later analysis.[6]

Between the seventh and ninth day after injection of the cholic acid-24-$^{14}$C, the animals were anesthetized with pentobarbital, and the gallbladder exposed through a ventral abdominal incision. Approximately 0.05 ml of bile was obtained by passing a 25-gauge needle through the liver into the gallbladder. The bile was diluted with saline and an aliquot immediately frozen for later analysis. The remainder (approximately 80%) was returned to the animal via a nasogastric tube.

Fourteen to 17 days after injection of cholic acid-24-$^{14}$C, the animals were again anesthetized; and 0.1–0.5 ml of bile was withdrawn from the gallbladder. The animals were killed by exsanguination, and any stones present in the gallbladder were removed, washed with distilled water, and weighed. The bile was frozen immediately and analyzed within 1–3 wk of collection.

### *Fecal Analysis*

Each fecal sample was homogenized in an Osterizer blender for 2 min at maximum speed. A 10 ml portion of the fecal homogenate, consisting of approximately 9 parts water to 1 part feces, was taken while the homogenate was being mixed and added to a preweight 125 ml glass-stoppered bottle. The bile acid fraction was isolated after removal of the neutral steroids as described previously.[9] An aliquot of the bile acid fraction was dissolved in 15 ml of Aquasol (New England Nuclear) and the radioactivity measured using a Beckman DPM-100 liquid scintillation spectrometer (Beckman Instruments Corp., Fullerton, Calif.). Quenching was corrected for by an external standard-channels ratio method, and the samples were counted to a 2-sigma error of less than 1%. The recovery of radiolabeled bile acids from the feces was greater than 98% as determined by measuring the radioactivity remaining in the fecal material after extraction of the bile acids.

Cholic acid-24-$^{14}$C recovery for the entire procedure was measured using two monkeys fed the mixed fat control diet. The animals were injected with the cholic acid-24-$^{14}$C, their feces were collected as described, and both animals were killed on the fourth day after isotope injection. An aliquot of bile was taken for analysis; and the remaining bile, as well as the gallbladder, liver, and the entire intestinal tract and its contents were digested with 10 N sodium hydroxide in 33% ethyl alcohol at 40–50°C for 48 hr. The radioactivity present in a weighed portion of this hydrolysate was determined as described for feces. Greater than 90% of the injected dose was recovered.

The fecal homogenate from a third monkey was divided into 20 separate, preweighed containers; and the contents of one container were analyzed with each set of determinations over a 2-mo period to measure the reproducibility of the fecal analysis procedure. The coefficient of variation of 4.6%.

### *Bile Analyses*

Bile samples were thoroughly mixed at room temperature, diluted with water, and the lipids extracted from an aliquot by the method of Folch et al.[7] One portion of the Folch lower phase was taken for determination to total lipid phosphorus by the method of Fiske and SubbaRow.[8] A second portion was dried under nitrogen, redissolved in 95% isopropanol, and the cholesterol concentration determined using the AutoAnalyzer II. Any bilirubin present was adsorbed to aluminum hydroxide prior to analysis.

A second aliquot of the diluted bile was deconjugated by the enzymatic procedure of Nair.[11] The free bile acids were extracted with ethyl ether and quantified by gas-liquid chromatography as described by Jones et al.[10] The bile acid analyses were done with and without the internal standard (7-ketolithocholic acid) to establish that interfering peaks did not appear in the area of the internal standard.

We compared total radioactivity present in the free bile acids to that present in the original bile aliquot to monitor the efficiency of the deconjugation process. In every instance, greater than 97% of the radioactivity was recovered as free bile acids after the deconjugation step.

### *Calculations*

The calculations involved in the bile acid kinetics were based on the assumption that the bile acid turnover conformed to a simplified first order kinetic model. The validity of this assumption has been discussed in detail by Lindstedt.[11,12]

The total isotope remaining in the body at any time was expressed as the fraction of the administered radioactivity remaining *(y)* and calculated from the equation

$$y = \frac{\mu\text{Ci injected} - \mu\text{Ci excreted}}{\mu\text{Ci injected}}. \quad (1)$$

A plot of $\log_e$ of $y$ versus time was subjected to statistical analysis to obtain the parameters of the first order rate equation

$$y = y_0 e^{-\alpha t}, \quad (2)$$

where $y_0$ is the fraction of the initial dose remaining at time $t = 0$, and the fractional clearance rate (FCR = $\alpha$) is the slope of the regression line. The line obtained was used to estimate the amount of radioactivity remaining in the animal at the time the bile sample was taken. The cholic acid (CH) specific radioactivity and the total radioactivity remaining were used to calculate the CH mass from the equation

$$\text{CH mass} = \frac{\text{Total radioactivity remaining (dpm)}}{\text{CH specific radioactivity (dpm/mg)}}. \quad (3)$$

The CH turnover rate was obtained from the equation

$$\text{Turnover rate (mg/day)} = \text{FCR} \times \text{CH mass}.$$

The mass of the chenodeoxycholic acid (CDC) pool was calculated using the equation

$$\text{CDC mass} = \text{CDC/CH mass ratio} \times \text{CH mass}.$$

The CDC/CH mass ratio used was the mean of that determined from the surgical and terminal bile samples. The ratio of the mass of CDC to CH varied by an average of less than 2% in bile from the same animal collected at the two time periods.

At the time the terminal bile sample was taken, 90%–95% of the administered radioactivity had been cleared from most of the animals. As a result, we were unable to obtain accurate measurements of the cholic acid pool size at that time.

The lithogenicity of each bile sample was expressed as the lithogenic index (the ratio of the amount of cholesterol actually present in a bile sample to the maximum amount of cholesterol that bile sample is capable of holding)[13] and calculated as suggested by Thomas and Hofmann.[14] We calculated the lithogenic index using the maximum cholesterol solubility as defined by Admirand and Small[15] as well as that described by Hegardt and Dam,[16] and Holzbach et al.[17]

### *Statistics*

The data were analyzed using a simple one-way analysis of variance, and individual comparisons were tested for significance using Duncan's new multiple range test.[18]

## RESULTS

### *Plasma Cholesterol Concentrations*

Table 2 shows mean plasma cholesterol concentrations and mean body weights for each dietary group for the entire 15–17 mo experimental period. There were no significant differences in body weight among the groups. Cholesterol feeding caused a significant ($p < 0.05$) increase in the variability in plasma cholesterol concentrations within groups, with the animals fed the butter diet showing the greatest individual variation, and the safflower oil and control groups, the least. After making the appropriate correction for nonhomogeneous variance, we found that the plasma cholesterol concentrations of the animals fed coconut oil were not significantly different from those of the butter-fed group, although the cholesterol

**Table 2. Mean Body Weights, Plasma Cholesterol Concentrations, and Gallstone Frequency in Squirrel Monkeys Fed Four Types of Dietary Fat**

| Diet Group | N | Body Weight (g) | Plasma* Cholesterol (mg/dl) | Gallstone† Frequency |
|---|---|---|---|---|
| Control | 7 | 859 ± 37‡ | 201 ± 10 | 0/7 |
| Coconut oil | 7 | 755 ± 51 | 373 ± 30 | 1/7 |
| Butter | 7 | 731 ± 67 | 444 ± 56 | 3/7 |
| Safflower oil | 10 | 721 ± 48 | 262 ± 10 | 9/10 |

*Mean plasma cholesterol concentration for the entire 15–17 months of the experiment.

† Number of animals with gallstones/number of animals examined.

‡Mean ± SEM.

concentrations of both groups were significantly higher ($p < 0.01$) than those of the animals fed safflower oil or control diets.

*Gallstones*

Gallstone frequency is shown in Table 2, and the stone numbers and stone weights for individual animals are contained in Table 3. The gallstone frequency at 17 mo was considerably higher in monkeys fed the safflower oil diet than in those fed coconut oil or butter, although the latter group contained the two animals with the most severe disease. None of the monkeys fed the low cholesterol control diet developed gallstones.

The stones produced by the butter-and safflower oil-fed animals were similar in appearance to those described by Osuga et al.[4] They ranged in size within a given gallbladder from 2.5 mm in diameter to particles 100–200 μm or less in diameter. The latter particles appeared as a finely meshed sand. The larger stones had mul-

**Table 3. The Number and Weight of the Stones Contained in the Gallbladders of Monkeys Fed Three Types of Dietary Fat**

| | Monkey Number | Stone Number* | Stone Weight |
|---|---|---|---|
| Coconut oil | 2049 | 125† | 62 |
| Butter | 2072 | 300–400 | 126 |
| | 2077 | 300–400 | 227 |
| | 2081 | 55 | 76 |
| Safflower oil‡ | 2084 | 75 | 57 |
| | 2085 | 100 | 95 |
| | 2086 | 108 | 50 |
| | 2087 | 94 | 14 |
| | 2091 | 95 | 41 |
| | 2092 | 39 | 53 |
| | 2094 | 21 | 56 |
| | 2095 | 75 | 34 |

*Includes only well-formed stones at least 0.5 mm in diameter. The gallbladders of four monkeys (2072, 2077, 2085, and 2086) contained numerous tiny stones which appeared as fine sand in the gallbladder. These tiny stones contributed to the stone weight, but were not counted.

† These stones were completely different in appearance from those produced by the other animals and did not appear to be the result of the same process taking place in the other animals.

‡Does not contain one safflower oil-fed monkey with gallstones which died during the experiment but whose stone number and stone weight were not recorded.

berry surfaces suggesting that they were the result of agglomeration of smaller and intermediate- sized stones. Several of the larger stones were cracked open and, although some were hollow, all contained pigment near the core.

Only one animal fed the coconut oil-containing diet developed gallstones, and these were different in structure from those produced by the other animals. These stones were all approximately the same size (1 mm in diameter), multifaceted, and quartz-like, and did not appear to be the result of the same process as that taking place in the other animals. All gallstones analyzed contained greater than 90% cholesterol by weight.

*Bile Acid Metabolism*

Typical radioactivity decay curves for individual animals from each dietary group are shown in Fig. 1. Cholic acid FCR, pool size, chenodeoxycholic acid pool size, and total bile acid pool size for each animal are presented in Table 4. The data for the excretion of cholic acid-24-$^{14}$C were adequately described by a first order kinetic model in the majority of the animals with the correlation coefficients ranging from 0.95 to 0.99. The reproducibility of the cholic acid-24-$^{14}$C FCR was tested in four animals (one from each dietary group) by repeating the isotope injection-fecal collection procedure 2 mo after the first such experiment. The two FCRs for each animal were compared and found not to differ significantly.

At time $t = 0$, essentially 100% of the isotope should be present in the animal if the cholic acid-24-$^{14}$C equilibrates rapidly with, and is metabolically indistinguishable from, the animal's own bile acids, i.e., $y_0$ should approximate 1.0. This value averaged 1.04 in animals free of gallstones (Fig. 1A, 1B, and 1C) and 0.76 in animals that had gallstones (Fig. 1D).

The animals fed butter were found to have the highest cholic acid FCR ($p < 0.05$), while the coconut oil-fed group had the lowest mean FCR, lower even than that of the noncholesterol-fed controls, although this difference was not significant.

When compared to the control group, those animals fed cholesterol and the saturated fats (butter or coconut oil) had significantly larger ($p < 0.05$) cholic and chenodeoxycholic acid pools. The safflower oil group also showed a slight increase in both cholate and chenodeoxycholate pool size; however, only the latter was significantly different ($p < 0.05$) from that of the control group. The difference between the total pool size of the safflower oil group and that of the control group was not significant.

*Bile Composition*

Table 5 shows the mean bile acid distribution of each dietary group. Secondary bile acids were absent from the bile of most of the animals. Only monkeys #2060, 2085, 2092, and 2093 had appreciable quantities of deoxycholic acid in their bile, and it comprised from 3%–16% of the total bile acids. Lithocholic acid was never present in more than trace amounts. Cholesterol feeding caused a significant decrease ($p < 0.05$) in the CH:CDC ratio in the bile.

The average biliary concentrations of cholesterol, bile acids, and phospholipid expressed as mmoles per liter, as well as moles percent and the lithogenic indices for each dietary group are shown in Table 6. There were no significant differences among dietary groups when the data were expressed as mmoles per liter, with the

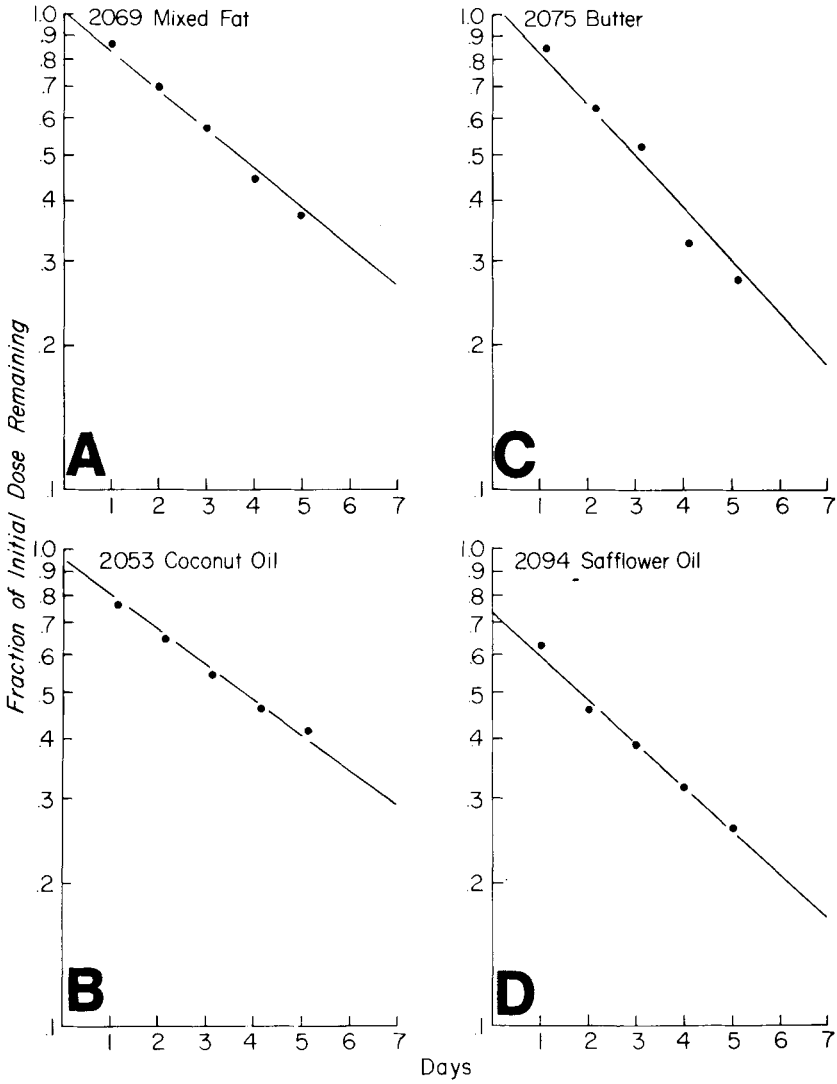


**Fig. 1. The rate of disappearance of cholic acid-24-$^{14}$C radioactivity from four squirrel monkeys fed (A) mixed fat, (B) coconut oil, (C) butter, and (D) safflower oil. The radioactivity remaining in the animal at any time was determined by subtracting the radioactivity excreted from that administered.**

exception that the bile acid concentration of the safflower oil group was significantly lower ($p < 0.05$) than that of the coconut oil-fed animals.

When the data were expressed in moles percent, the bile samples from the safflower oil group had the highest mean percentage of cholesterol and the lowest mean percentage of bile salt. Both components were significantly different ($p < 0.05$) from those of the butter and control groups. We were not able to show a difference in these components between the safflower oil- and coconut oil-fed groups.

Those monkeys consuming safflower oil had the highest mean lithogenic index regardless of whether the cholesterol solubility curve of Holzbach[17] or that of Ad-

**Table 4. Cholic Acid Fractional Clearance Rate, Pool Size, Chenodeoxycholic Acid Pool Size, and Total Bile Acid Pool Size in Squirrel Monkeys Consuming Different Dietary Fats***

| Monkey Number | Cholic Acid FCR | Cholic Acid Pool Size (mg) | Chenodeoxycholic Acid Pool Size (mg) | Total Bile Acid Pool Size (mg) |
|---|---|---|---|---|
| | | Control | | |
| 2060 | 0.159 | 44† | 8 | 52 |
| 2061 | 0.252 | 18 | 4 | 22 |
| 2063 | 0.313 | 36 | 5 | 41 |
| 2064 | 0.136 | 43 | 17 | 60 |
| 2067 | 0.202 | 40 | 8 | 48 |
| 2068 | 0.090 | 74 | 28 | 102 |
| 2069 | 0.204 | 85 | 40 | 125 |
| | 0.194± 0.028‡ | 49 ± 9 | 16 ± 5 | 64 ± 14 |
| | | Coconut Oil | | |
| 2049 | 0.067 | 86 | 30 | 116 |
| 2050 | 0.144 | 89 | 41 | 130 |
| 2051 | 0.134 | 70 | 30 | 100 |
| 2053 | 0.170 | 76 | 27 | 103 |
| 2057 | 0.110 | 97 | 39 | 136 |
| 2058 | 0.247 | 188 | 94 | 282 |
| | 0.145± 0.025 | 101 ± 18 | 44 ± 10 | 144 ± 28 |
| | | Butter | | |
| 2072 | 0.599 | 104 | 39 | 143 |
| 2075 | 0.266 | 40 | 66 | 106 |
| 2076 | 0.426 | 119 | 131 | 250 |
| 2081 | 0.234 | 151 | 110 | 261 |
| 2083 | 0.264 | 107 | 58 | 165 |
| | 0.358± 0.069 | 104 ± 18 | 81 ± 17 | 185 ± 30 |
| | | Safflower Oil | | |
| 2084 | 0.224 | 180 | 57 | 237 |
| 2085 | 0.116 | 44† | 19 | 63 |
| 2087 | 0.179 | 66 | 37 | 103 |
| 2091 | 0.158 | 69 | 35 | 104 |
| 2092 | 0.119 | 26† | 19 | 45 |
| 2093 | 0.339 | 48† | 18 | 66 |
| 2094 | 0.251 | 82 | 46 | 128 |
| | 0.198± 0.030 | 74 ± 19 | 33 ± 6 | 107 ± 24 |

*Bile acid turnover and pool size were measured in 25 animals. The turnover studies could not be completed in 3 animals due to incomplete fecal collections. These included one coconut oil-fed, one butter-fed, and one safflower oil-fed monkey. One safflower oil-fed monkey with gallstones developed nephrotic syndrome and a second safflower oil-fed monkey with gallstones died 4 mo prior to initiation of the turnover studies. Both were excluded. One butter-fed monkey with gallstones produced a cholic acid-24-$^{14}$C excretion curve that failed to reach a final exponential rate of decay and was also excluded.

† Includes deoxycholic acid mass.

‡ Mean ± SEM.

**Table 5. The Biliary Bile Acid Distribution in Squirrel Monkeys Consuming Different Dietary Fats**

| Dietary Fat | Cholate | Chenodeoxycholate | Deoxycholate |
|---|---|---|---|
| Control | 78.1 ± 2.8 | 21.5 ± 2.9 | 0.4 ± 0.4 |
| Coconut oil | 68.4 ± 2.4 | 31.6 ± 2.4 | ND* |
| Butter | 61.1 ± 5.8 | 38.9 ± 5.8 | ND |
| Safflower oil | 60.9 ± 3.5 | 35.4 ± 2.9 | 3.6 ± 2.0 |

Results are expressed as percentage of total biliary bile acids (mean ± SEM). Bile from animals fed the low cholesterol control diet had significantly ($p < 0.05$) greater percentage of bile acids as cholic acid and less as chenodeoxycholic acid than any of the other dietary groups.

*ND = none detected.

mirand and Small[15] was used to calculate bile lithogenicity. This difference was significant in all but one instance, that being the comparison between the butter and safflower oil groups using Holzbach's line of maximum cholesterol solubility.

The majority of the points representing bile compositions clustered about Holzbach's line of maximum cholesterol solubility regardless of the dietary group examined. In no case did a point exceed the cholesterol solubility curve of Admirand and Small.

## DISCUSSION

We have reported previously[3] that squirrel monkeys, fed diets containing safflower oil and cholesterol, have more cholelithiasis than do their counterparts fed cholesterol in combination with butter or coconut oil. In the present study, we confirm this observation and suggest that the increased frequency of gallstones among animals fed safflower oil is associated with a decreased capacity to expand their bile acid pools.

The conclusion that safflower oil-fed monkeys develop more gallstones than those fed butter was based on the frequency of occurrence of gallstones among the animals of each dietary group after 17 mo. The probability of obtaining frequency ratios (9/10 for the safflower oil group versus 3/7 for the butter group and 1/7 for the coconut oil group) this different simply by chance is less than 10% (Fisher Exact Test). These data support the concept that Safflower oil-containing diets stimulate stone formation more rapidly than diets containing butter or coconut oil. A similar conclusion has been reached regarding the influence of safflower oil-and butter-containing diets on cholelithiasis in African green monkeys.[19] We should

**Table 6. Cholesterol, Phospholipid, and Bile Salt Composition and Lithogenic Index of Bile From Squirrel Monkeys Consuming Different Dietary Fats**

| | mMoles Per Liter | | | Moles Percent | | | Lithogenic Index | |
|---|---|---|---|---|---|---|---|---|
| Dietary Fat | Cholesterol | Phospholipid | Bile Acid | Cholesterol | Phospholipid | Bile Acid | Admirand and Small[15] | Holzbach[17] |
| Control | 9.3 | 36.7 | 169.1 | 4.2 | 16.6 | 79.3 | 0.45 | 0.77 |
| | ±1.8 | ±7.5 | ±33.2 | ±0.5 | ±1.9 | ±2.2 | ±0.05 | ±0.09 |
| Coconut oil | 17.4 | 55.4 | 254.5 | 4.8 | 15.9 | 79.4 | 0.51 | 0.85 |
| | ±5.0 | ±15.6 | ±49.9 | ±0.9 | ±2.6 | ±3.5 | ±0.08 | ±0.11 |
| Butter | 14.1 | 45.8 | 199.6 | 4.9 | 15.5 | 79.6 | 0.52 | 0.93 |
| | ±5.2 | ±16.2 | ±54.3 | ±0.5 | ±1.5 | ±1.9 | ±0.05 | ±0.08 |
| Safflower oil | 10.8 | 28.4 | 122.0 | 6.9 | 18.3 | 74.5 | 0.71 | 1.16 |
| | ±1.9 | ±3.7 | ±10.2 | ±0.8 | ±1.6 | ±1.8 | ±0.07 | ±0.09 |

Results are expressed as mean ± SEM. The lithogenic index was calculated as proposed by Thomas and Hoffman[14] using the curves of maximum cholesterol solubility described by Admirand and Small[15] and Holzbach.[17]

point out that continued feeding of these diets will eventually produce a high gallstone frequency in the butter group as well, but feeding periods of up to 2 yr are required.

Although only 3 of the 7 butter-fed animals developed cholelithiasis, the 2 most severely affected monkeys (2072 and 2077) were in this group. We have no explanation for this observation other than to point out that a higher average gallstone weight in the butter-fed group is contrary to our previous observations.[3]

Because of the small quantities of bile contained in the gallbladder, the difficulty encountered in attempting to pass a duodenal cannula and the relatively large dead space of the cannula, we were unable to utilize the standard techniques of Lindstedt[12] to measure bile acid kinetics in the squirrel monkey. As a result, we devised the procedure described here rather than to attempt to sample the bile directly. This procedure has the advantage of permitting the measurement of the various parameters of bile acid metabolism with minimum amounts of manipulation of the animals during the period of study. For this technique to be considered valid, however, the animals must be in a steady state with regard to bile acid synthesis and excretion, and the clearance of bile acid in the feces must be regular and fairly rapid. Both of these criteria appear to have been met in this study. The body weights and plasma cholesterol concentrations had stabilized well before the bile acid kinetic studies were begun, and it is reasonable to assume that the bile acid pools had also reached a steady state by this time.

Fecal flow markers were not used in this study since squirrel monkeys fed similar diets have been shown to have fecal transit times considerably less than 24 hr.[6] The goodness of fit of the data to the first order kinetic model, with $y$-intercepts averaging about 1.0 is a clear indication that the fecal flow was rapid and regular. If the transit time did exceed 24 hr, such that larger quantities of the isotope were excreted on days 2 and 3 after injection than on day 1, the resulting curves would have $y$-intercepts exceeding 1.0. This did not occur.

Nine animals had fecal excretion curves with $y$-intercepts of less than 0.9. Eight of these nine animals had gallstones. This suggests that the gallstones prevented complete emptying of the gallbladder in these animals, thereby leaving a residual pool of bile acids that failed to equilibrate initially with the isotope moving through the rest of the enterohepatic circulation. As a result, the fraction of the isotope lost from the enterohepatic circulation within the first few hours after injection was greater than the actual cholic acid fractional clearance rate. Even in these animals, however, the majority of the isotope eventually equilibrated with the bile acid pool, for in every case the isotope died away as a single exponential function after 24 hr. Beyond that time it is reasonable to assume that the isotope lost in a given time interval was equal to the fraction of the cholic acid pool turning over during that interval. It should be pointed out that the problem of slow mixing of the injected isotopically labeled bile acid in individuals with existing cholelithiasis is not unique to this method, but also would occur with the traditional Lindstedt method. In every study, with which we are familiar, in which bile acid metabolism was measured by the Lindstedt technique, the investigators failed to sample the bile prior to 24 hr after isotope administration. As a result, they would not have detected the slow-mixing phenomenon noted here.

Our estimates of bile acid pool size are somewhat larger than those reported by Osuga[21] and Tanaka et al.[22] for squirrel monkeys. Several pieces of evidence sug-

gest that these are real differences and not due to some unrecognized methodologic difficulties.

Recovery of the injected dose in two control animals exceeded 90%, indicating that losses of isotope other than in the feces were minimal. In addition, the coefficient of variation among 20 replicate analyses was 4.6%, indicating considerable precision in repeated measurements of fecal radioactivity. We have also measured bile acid excretion rate in monkeys fed similar diets using sterol balance techniques,[20] and the values obtained for excretion were remarkably similar to those reported here. These findings, combined with the fact that the excretion of the $^{14}C$-radioactivity in the feces was first order, leads us to conclude that the estimate of cholate excretion obtained by the fecal isotopic turnover method is both accurate and precise in this species.

The most logical explanation for the differences in bile acid pool size between the present study and that of Osuga et al.[21] and Tanaka et al.[22] relates to the differences in the composition of the diets. The diets used by Osuga and Tanaka were semisynthetic diets, whereas our diets were composed largely of natural products. An example of the potent effect the type of diet can have on bile acid pool size was noted by Portman and Osuga[23] who found that the primary bile acid pool size in monkeys fed a commercially available chow was more than twice as large as that in monkeys fed semisynthetic diets. The average size of the bile acid pool in the squirrel monkeys studied by Portman and fed the commercial diet was approximately 90 mg. This was not markedly different from a mean of 120 mg, obtained by averaging total bile acid pool size from the data of all animals described in Table 3, even though diets were different in these two studies. The same is true of the cholic acid fractional catabolic rate that averages 0.17 in squirrel monkeys fed the commercial diet as described by Portman and Osuga[23] versus 0.22 in those fed the diets used in the present study. Thus, taking into consideration the potential influence of diets of different composition on bile acid pool size, it is not surprising that the bile acid pool sizes, determined from the animals in this study, are not identical to those reported by others using different diets and experimental conditions.

It seems clear that the development of gallstones in cholesterol fed squirrel monkeys is the result of cholesterol "overloading," i.e., an increase in the flux of cholesterol through the biliary system so great that it exceeds the animal's capacity to maintain it in miceller solution. If this is the case, three possibilities exist that could account for the differences in stone frequency among the various diet groups after 15–17 mo.

First, absorption of dietary cholesterol could be greater in the safflower oil group than the other diet groups, thereby causing this group to "overload" faster. However, previous studies from our laboratory have indicated that the absorption of cholesterol in squirrel monkeys is not enhanced by the isocaloric substitution of safflower oil for butter.[20]

A second possibility is that the butter- and safflower oil-fed monkeys absorb the same amounts of cholesterol, but that safflower oil causes more of the newly absorbed cholesterol to be channeled into the bile as opposed to other body pools. This could also account for the lower plasma cholesterol values in the latter group. We cannot rule out this possibility absolutely; however, if safflower oil does stimulate cholesterol secretion in the bile, one might expect the whole body pools of

cholesterol to be larger in butter-fed animals than in those fed safflower oil; and this is not the case.[24]

The third explanation for the increased stone frequency among the animals fed safflower oil, and the one our data supports, is that butter-fed animals were able to stimulate bile acid synthesis in response to dietary cholesterol to a greater degree than their counterparts fed safflower oil. This would not only result in a reduced cholesterol flux through the bile (since cholesterol is the substrate for bile acid synthesis), but also increase the cholesterol solubilizing capacity of the bile.

Butter-fed animals expanded their pool of bile acids threefold compared to cholesterol-free controls, while safflower oil-fed animals showed only a slight and statistically insignificant expansion of their bile acid pool. In addition, the cholic acid fractional clearance rate in butter-fed animals was approximately double that of safflower oil-fed animals. These findings, taken together, support the conclusion that bile acid flux through the enterohepatic circulation was considerably greater in butter-fed animals than in safflower oil-fed animals. Grundy[25] noted a similar increase in the cholesterol-to-bile acid ratio of bile in human beings when hypertriglyceridemic individuals were switched from lard to safflower oil.

Coconut oil was the least lithogenic of the dietary fats tested. To date we have examined 24 squirrel monkeys fed this diet and only 1 had gallstones. Even though the average bile acid pool of the coconut oil-fed animals was larger than that of controls, the estimated fractional catabolic rate was the lowest of any group studied. Hence, it would not appear that a high rate of bile acid flux through the enterohepatic circulation was responsible for the lack of cholelithiasis in these animals. Previous studies[24] have shown that coconut oil-fed monkeys had the smallest body pools of cholesterol of any of the diet groups and indicated that cholesterol input from absorption and synthesis was also reduced in these animals. These data suggest that coconut oil-fed monkeys do not absorb cholesterol as efficiently as the animals in the other diet groups and that the lack of gallstones in the monkeys fed coconut oil is due to a relatively low rate of cholesterol flux through the biliary system secondary to poor cholesterol absorption.

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## APPENDIX

Only four animals had appreciable quantities of deoxycholic acid in their bile. If the cholic acid radioactivity excretion was first order, cholic acid served as sole precursor of deoxycholic acid, and deoxycholic acid was not converted to cholic acid, then the fraction of the injected radioactivity remaining in these animals ($y_I$) at any time ($t$) is described by the equation

$$y_I = y_{CH} + y_D,$$

where $y_{CH}$ is the fraction of the injected radioactivity remaining as cholic acid-24-$^{14}$C at any time and is obtained from the equation

$$y_{CH} = e^{-(k_1 + k_2)t},$$

and $y_D$ is the fraction of the injected radioactivity remaining as deoxycholic acid-24-$^{14}$C at any time and is obtained from the equation

$$y_D = \frac{k_3}{k_3 - (k_1 + k_2)}\left(\frac{B}{A}\right)\left(y_{CH} - e^{-k_3 t}\right),$$

where $k_1$ is the fraction of cholic acid-24-$^{14}$C converted to deoxycholic acid-24-$^{14}$C per day, $k_2$ is the fraction of cholic acid-24-$^{14}$C radioactivity excreted per day, $k_3$ is the fraction of deoxycholic acid-24-$^{14}$C radioactivity excreted per day, and $B/A$ is the deoxycholic acid:cholic acid ratio in the bile.

As $t \rightarrow \infty$, a plot of $\log_e y_1$ versus $t$ becomes linear, after which the slope of the line is equal to the cholic acid FCR.

A least-squares fit of a line through the last three points of the experimental curve was used to estimate the cholic acid FCR and fraction of injected radioactivity remaining at the time of the bile sample in these four animals. The pool size, defined as the cholic acid + deoxycholic acid mass, was obtained from the equation

$$\text{Pool size} = \frac{\text{Radioactivity remaining (dpm)}}{12\alpha\text{-hydroxylated bile acid specific activity (dpm/mg)}}.$$

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